

Table 4: The Truth Table for the Exclusive Or of Two Propositions

p	q	$p \oplus q$
T	T	F
T	F	T
F	T	T
F	F	F

Example 8:

Let p and q be the propositions that state "A student can have a salad with dinner" and "A student can have soup with dinner," respectively. What is $p \oplus q$, the exclusive or of p and q ?

Soln: A student can either have salad or soup with dinner, but not both.

Conditional Statements

Definition 5: Let p and q be propositions. The conditional statement $p \rightarrow q$ is the proposition "if p , then q ". The conditional statement $p \rightarrow q$ is false when p is true and q is false, and true otherwise. In the conditional statement $p \rightarrow q$, p is called the hypothesis (or antecedent or premise) and q is called the conclusion (or consequence).

The statement $p \rightarrow q$ is called a conditional statement because $p \rightarrow q$ asserts that q is true on the condition that p holds. A conditional statement is also called an implication.

Table 5: The Truth Table for the Conditional Statement $p \rightarrow q$

p	q	$p \rightarrow q$
T	T	T
T	F	F
F	T	T
F	F	T

A variety of terminology is used to express $p \rightarrow q$

$p \rightarrow q$	" p implies q "
"if p , then q "	" p only if q "
"if p , q "	"a sufficient condition for q is p "
" p is sufficient for q "	" q whenever p "
" q if p "	" q is necessary for p "
" q when p "	" q follows from p "
"a necessary condition for p is q "	" q provided that p "
" q unless $\neg p$ "	